

Functions and transformations - Set 2 - Solutions

Specification MM1.7, MM8.2, MM8.3, MM8.4.

Answer key

Question	Answer	Topic
1	D	Functions and transformations
2	A	Functions and transformations
3	F	Functions and transformations
4	F	Functions and transformations
5	D	Functions and transformations
6	F	Functions and transformations

Worked solutions

Question 1 FUN-04 · Functions and transformations · spec MM1.7

Which of the following functions, each with domain the set of all real numbers, is one-to-one?

Answer: D

$x^3 + x$ is strictly increasing (its derivative $3x^2 + 1 > 0$), so different inputs give different outputs. $x^3 - x$ takes the value 0 at $x = -1, 0, 1$; the others are even or periodic.

Question 2 FUN-07 · Functions and transformations · spec MM8.2

The function f satisfies $f(x) = x^2$ for all x . Which of the following transformations maps $y = f(x)$ onto $y = 4x^2 - 8x + 4$?

Answer: A

$4x^2 - 8x + 4 = 4(x - 1)^2$, i.e. $y = 4f(x - 1)$: translate 1 right, then stretch vertically by 4. (Equivalently $y = f(2x - 2) = f(2(x - 1))$: stretch parallel to the x -axis by factor $\frac{1}{2}$ then translate 1 right - not offered.)

Question 3 FUN-10 · Functions and transformations · spec MM8.2

$f(x) = x^2 + 1$. Which of the following statements is/are true?

- I $f(f(x)) = x^4 + 2x^2 + 2$ for all x
- II $f(x + 1) = f(x) + 1$ for all x
- III $f(2x) = 4f(x) - 3$ for all x

Answer: F

I: $f(f(x)) = (x^2 + 1)^2 + 1 = x^4 + 2x^2 + 2$: true. II: $f(x + 1) = x^2 + 2x + 2 \neq x^2 + 2$: false. III: $f(2x) = 4x^2 + 1$ and $4f(x) - 3 = 4x^2 + 1$: true.

Question 4 FUN-12 · Functions and transformations · spec MM1.7

Let $f(x) = \sqrt{x - 3}$ for $x \geq 3$ and $g(x) = x^2 + 3$. Which of the following statements is/are true?

- I $g(f(x)) = x$ for all $x \geq 3$
- II $f(g(x)) = x$ for all real x
- III $f(g(x)) = |x|$ for all real x

Answer: F

I: $g(f(x)) = (\sqrt{x - 3})^2 + 3 = x$ for $x \geq 3$: true. $f(g(x)) = \sqrt{x^2} = |x|$, so II fails (e.g. $x = -1$) and III is true.

Question 5 FUN-16 · Functions and transformations · spec MM1.7

The function f is defined by $f(x) = |x - 2| + |x + 2|$ for all real x . What is the minimum value of $f(x)$, and for how many values of x is it attained?

Answer: D

For $-2 \leq x \leq 2$: $f(x) = (2 - x) + (x + 2) = 4$. Outside this interval $f(x) > 4$. So the minimum 4 is attained on the whole interval $[-2, 2]$.

Question 6 X2-11 · Functions and transformations · spec MM8.2

Let $f(x) = x^2$ for all real x , and let g be the function whose graph is obtained by translating the graph of $y = f(x)$ by 2 units in the positive x -direction and then reflecting in the y -axis. Which of the following statements is/are true?

I $g(x) = (x + 2)^2$

II $g(x) = (x - 2)^2$

III The graph of g has its vertex at $(-2, 0)$

Answer: F

Translation gives $(x - 2)^2$; reflecting in the y -axis replaces x by $-x$: $(-x - 2)^2 = (x + 2)^2$, with vertex at $(-2, 0)$.